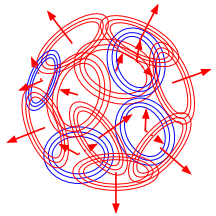
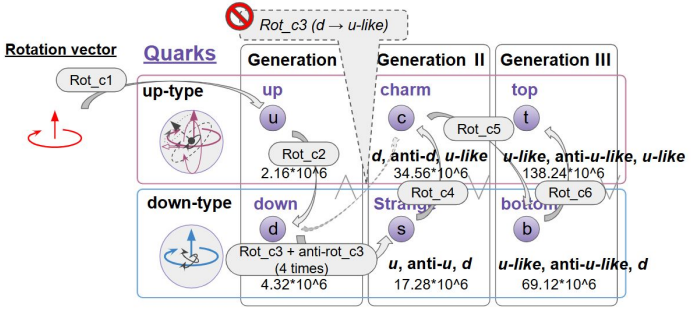
R-space Theory 1.0

- Geometric Integration of Relativity and Quantum Mechanics -



Chap.8: The Mechanism of Quark Generations



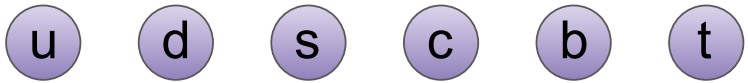
1. What are Quark Flavors?

In addition to up quarks and down quarks, which are the elementary particles that make up ordinary matter such as protons and neutrons, there are quarks that only appear in high-energy environments such as cosmic rays and accelerator experiments.

Including these, six types of quarks are currently known.

These types are called flavors: up (*u*), down (*d*), strange (*s*), charm (*c*), bottom (*b*), and top (*t*).

Each flavor differs in its charge, mass, behavior in weak interactions, lifespan, and decay pathways.



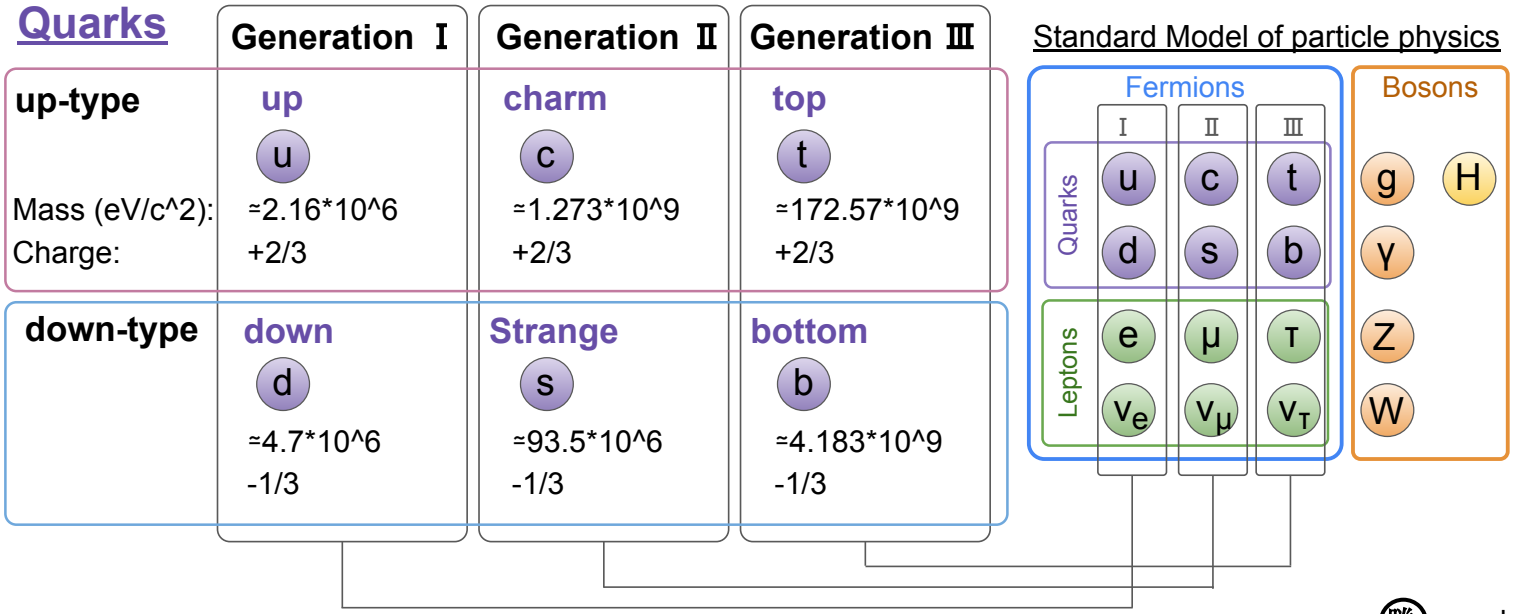
1. What are Quark Flavors?

Furthermore, quarks are divided into three generations, with each generation consisting of a pair of an up-type (+2/3 charge) and a down-type (-1/3 charge) quark.

Quarks	Generation Ⅰ	Generation Ⅱ	Generation Ⅲ
up-type	up u	charm c	top t
Mass (eV/c^2):	≈2.16*10^6	≈1.273*10^9	≈172.57*10^9
Charge:	+2/3	+2/3	+2/3
down-type	down d	Strange s	bottom b
	≈4.7*10^6	≈93.5*10^6	≈4.183*10^9
	-1/3	-1/3	-1/3

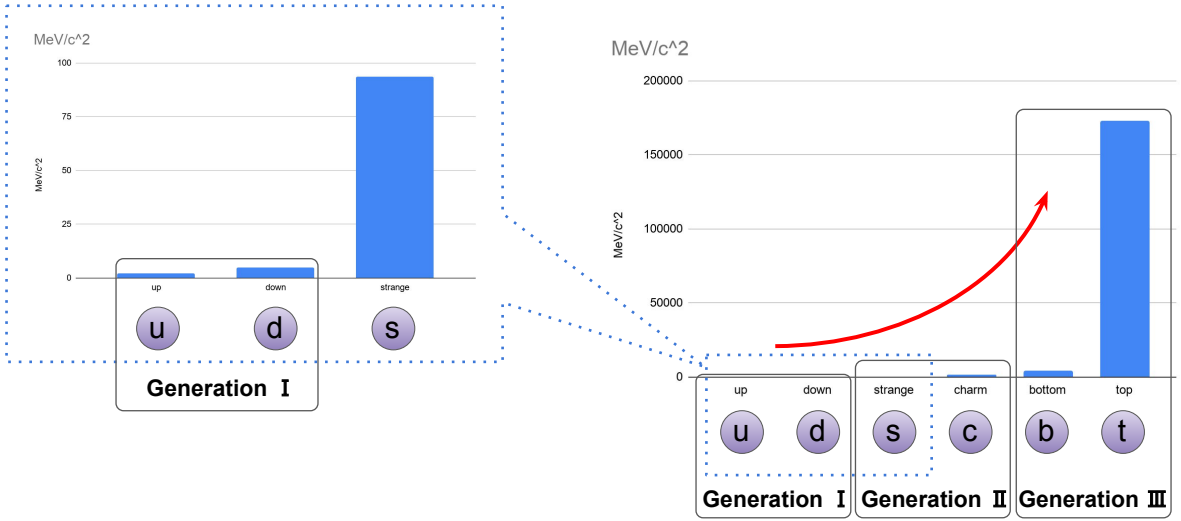
1. What are Quark Flavors?

The three generations of quarks completely parallel the three generations of leptons (electron, muon, tau, and their respective neutrinos), a structure known as the generation structure.



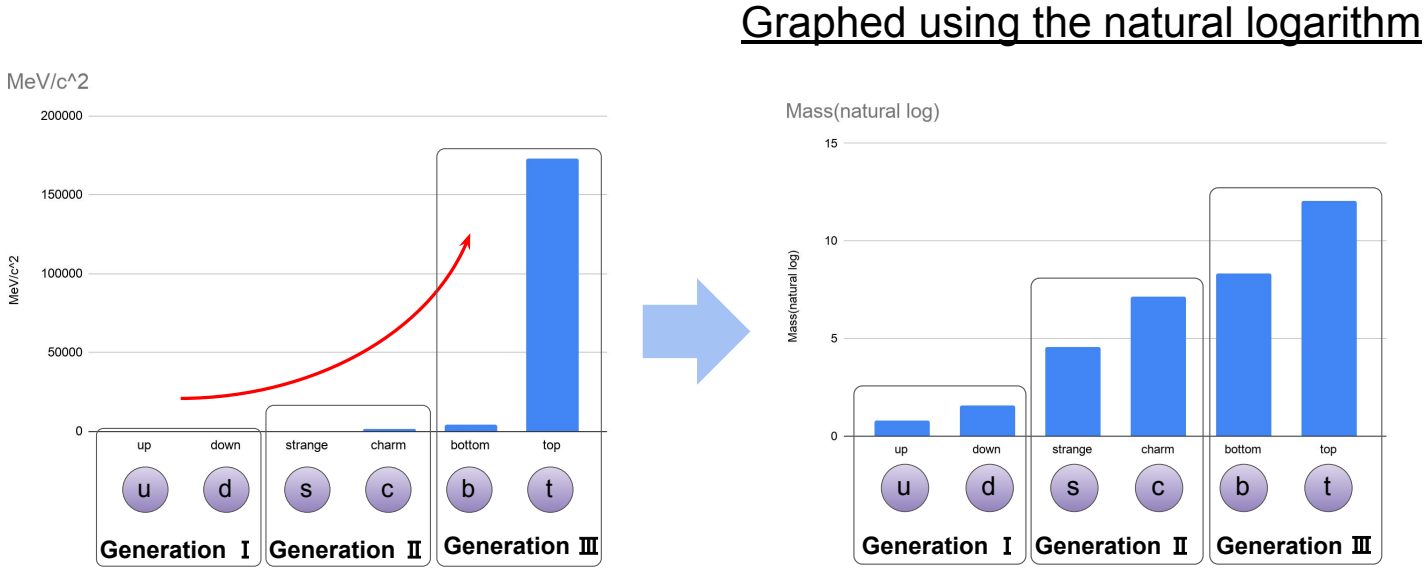
1. What are Quark Flavors?

The masses increase exponentially with each generation: the up and down quarks are on the order of a few MeV/c²,the charm and strange quarks range from several hundred MeV/c² to the GeV/c² scale,and the top and bottom quarks lie in the multi-GeV region, with the top quark reaching about 173 GeV/c².



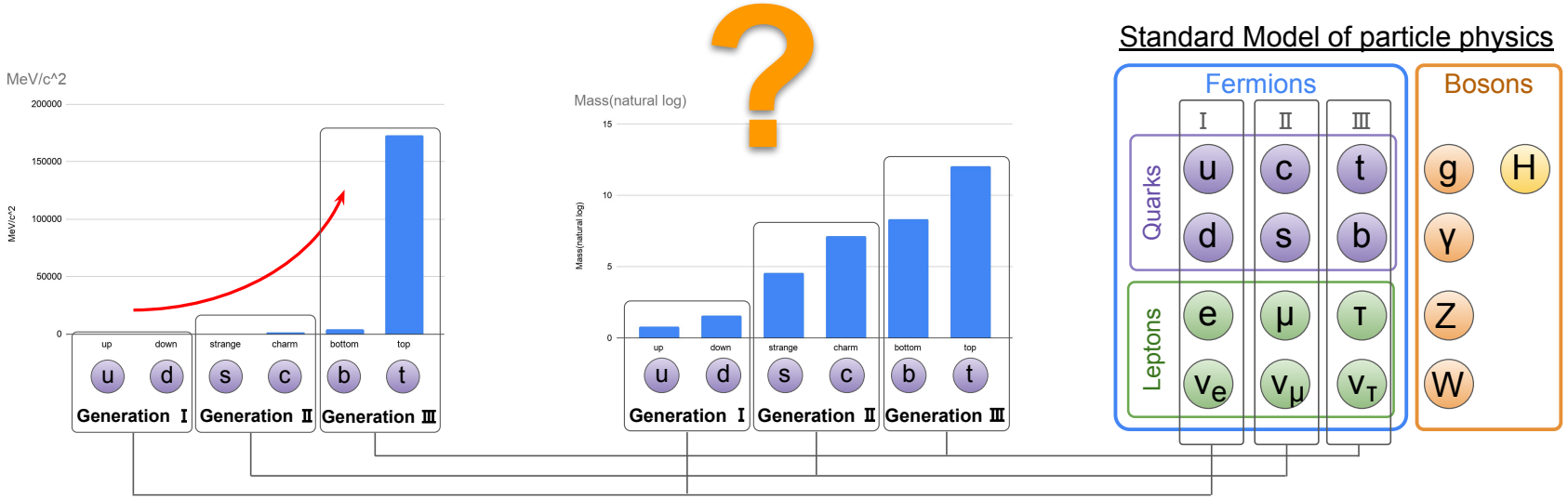
1. What are Quark Flavors?

When the mass of each flavor is graphed using a natural logarithm, it becomes clear that they increase in a roughly step-like fashion as the generations rise.



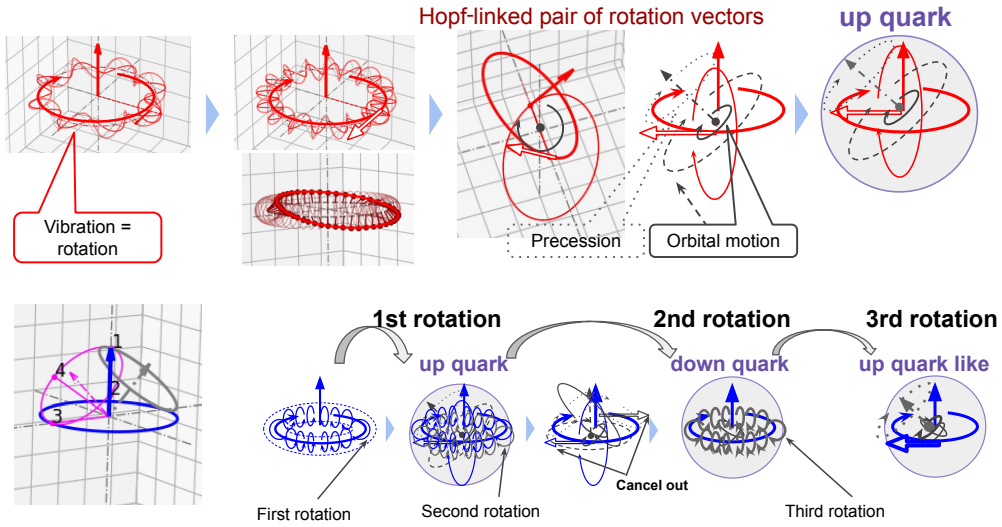
1. What are Quark Flavors?

In physics, the mystery of why quark masses are scattered so neatly yet across such a wide range is called the "Flavor Problem," and research on this continues today.



2. Rotation Vectors and Mass

In Chapter 7, it was explained that if the energy of the rotation vector (angular momentum) constituting a quantum is high, the phase circle vibrates (the angular momentum vibrates), and this vibration becomes the rotation of color charge, causing the rotation vector to change between up-type and down-type.



2. Rotation Vectors and Mass

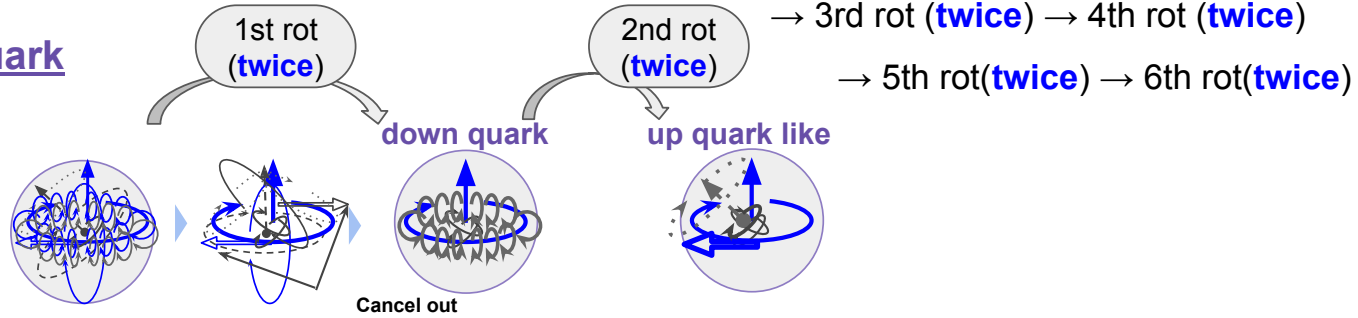
The same logic can be applied to strange, charm, bottom, and top quarks.

Starting from the up quark mass of approximately 2.16 MeV/c², the state is rotated between up-type and down-type.

Assuming that rotating the state requires energy equal to the mass before rotation, and since mass = energy, the mass of the quark after the state rotates becomes twice the mass of the quark before rotation.

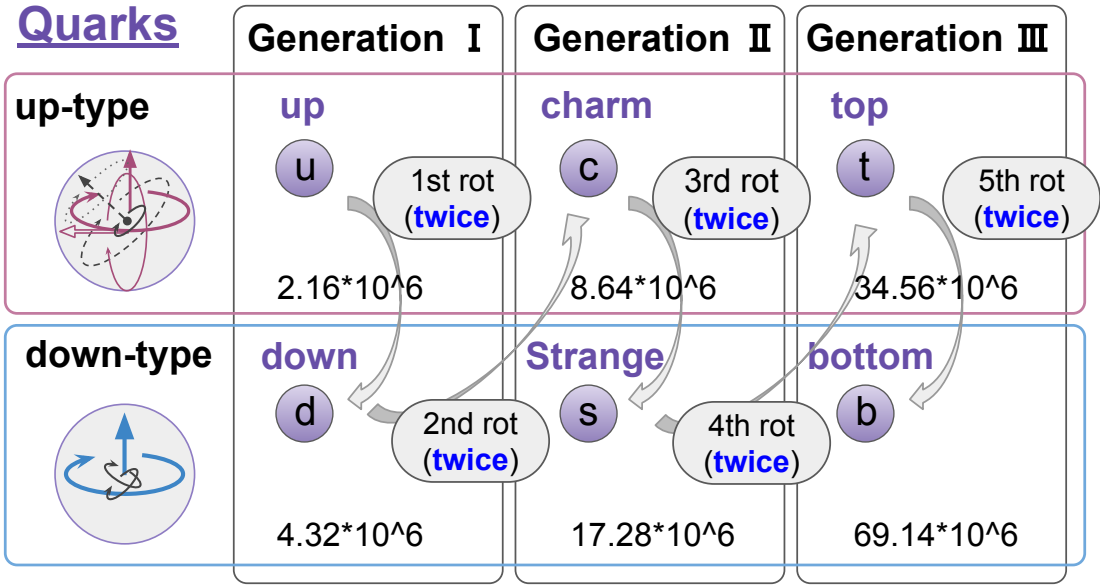
Starting from up quark

2.16 MeV/c²

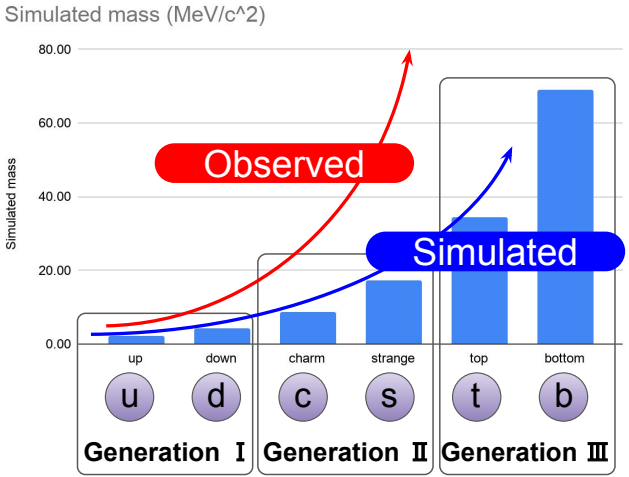


2. Rotation Vectors and Mass

While the calculated mass increase in this simulation is smaller than the observed data, it successfully explains the exponential increase in mass.



Simulated mass

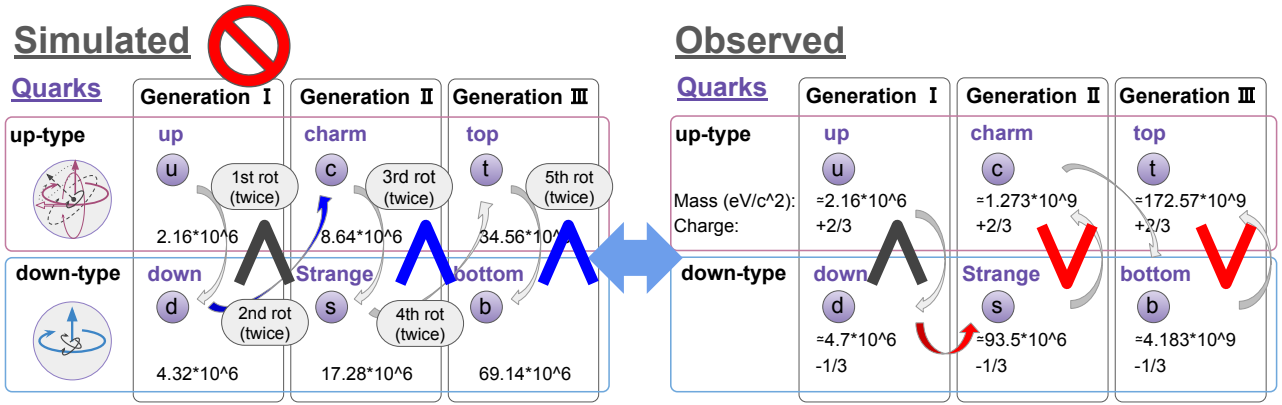


2. Rotation Vectors and Mass

This difference in increase rates might be explained by changing assumptions, such as the effect of the orbital revolution of quarks within hadrons.

However, this simulation has a challenge: in the 2nd and 3rd generations, the mass relationship between up-type and down-type quarks is inverted compared to actual observation.

This inversion of mass by generation is related to antiparticles.



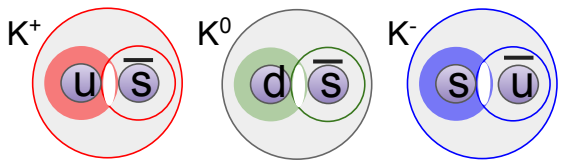
3. Quark Decay Patterns

According to the Standard Model, the **down quark** transforms into an **up quark** through the mediation of a W^- boson, and the W^- subsequently decays into an electron and an electron antineutrino ($d \rightarrow u\ W^-$, $W^- \rightarrow e^-\ anti\text{-}\nu_e$).

This is the pattern observed in β decay.

(In R-space Theory, as explained in the previous chapter, the pre-decay energy is considered to be distributed among all three quarks rather than being carried solely by the down quark.)

The next heavier quark, the **strange quark**, appears in kaons (K^+ : *u anti-s*, K^0 : *d anti-s*, K^- : *s anti-u*).



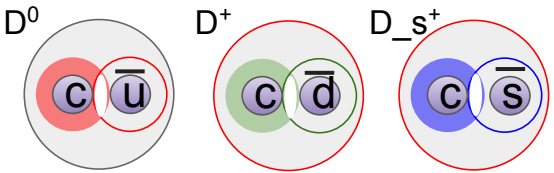
3. Quark Decay Patterns

Although the strange quark has several decay channels, the dominant mode is an initial transition into an up quark and a W^- boson ($s \rightarrow u\ W^-$).

The W^- boson then decays into a quark pair—such as a π^- meson (*d anti-u*), a K^- meson (*s anti-u*)—or into a lepton pair such as an electron and an electron antineutrino.

Depending on the parent hadron containing the strange quark, the W^- boson most frequently decays into hadrons (π or K).

The **charm quark** is found in D mesons (D^0 : *c anti-u*, D^+ : *c anti-d*, D_s^+ : *c anti-s*).



3. Quark Decay Patterns

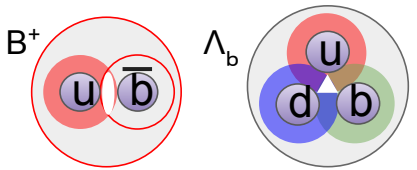
It primarily decays into strange or down quarks, with the transition ($c \rightarrow s$) being the most probable.

The main decay channels are "$c \rightarrow s\ W^+$, $c \rightarrow d\ W^+$, $c \rightarrow u\ W^+$", with $c \rightarrow s\ W^+$ dominating.

The W^+ boson decays into lepton pairs or quark pairs such as "*u anti-d*, *u anti-s*, *c anti-d*, *c anti-s*".

The **bottom quark** appears in *B* mesons (B^+ : *u anti-b*, B^0 : *d anti-b*, Bs^0 : *s anti-b*, Bc^+ : *c anti-b*) and in the Λ_b baryon (*u d b*).

Its main decay channels are "$b \rightarrow c\ W^-$, $b \rightarrow u\ W^-$" .



3. Quark Decay Patterns

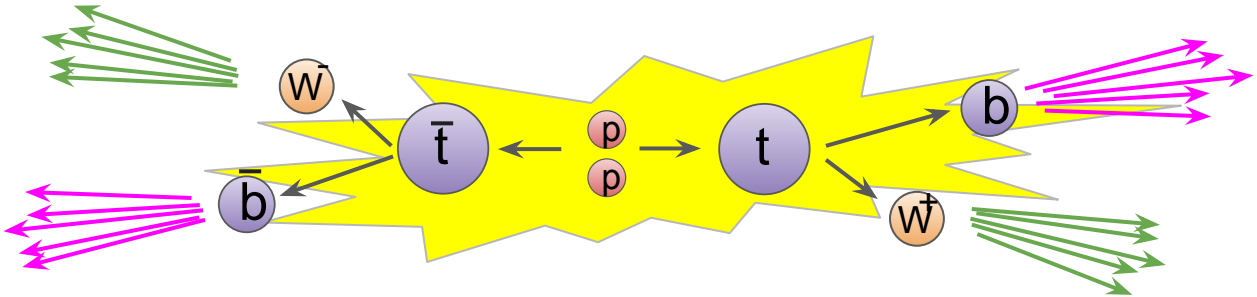
The W^- boson undergoes a variety of weak decays, including semileptonic and rare decay modes.

The **top quark** does not hadronize because its lifetime is far too short.

It was discovered through analyses of b-jets (bundles of hadrons produced by bottom-quark hadronization) and leptonic decays of the W boson.

The top quark decays almost exclusively via $t \rightarrow b\, W^+$.

The W^+ boson then decays into quark pairs or lepton pairs.



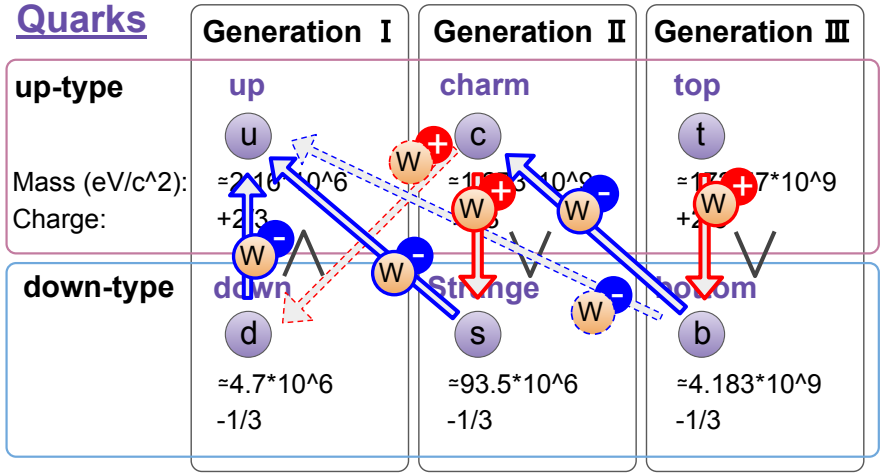
3. Quark Decay Patterns

Summary of the decay patterns are as following.

- $d \rightarrow u\ W^-$, $W^- \rightarrow e^-\ anti\text{-}\nu_e$
- $s \rightarrow u\ W^-$, $W^- \rightarrow \pi^-$, K^-, or lepton pairs
- $c \rightarrow s\ W^+$, $c \rightarrow d\ W^+$, $W^+ \rightarrow$ lepton pairs or quark pairs
- $b \rightarrow c\ W^-$, $b \rightarrow u\ W^-$, $W^- \rightarrow$ weak, semileptonic, or rare decays
- $t \rightarrow b\ W^+$, $W^+ \rightarrow$ quark pairs or lepton pairs

3. Quark Decay Patterns

The relationship can be illustrated in the following diagram, where decay channels with high probability are shown with solid arrows, and those with lower probability are indicated with dashed arrows.



3. Quark Decay Patterns

Focusing on the dominant decay modes, one finds a clear pattern in the second and third generations:

up-type quarks tend to decay into the down-type quark of the same generation, whereas down-type quarks preferentially decay into up-type quarks across generations.

For the first generation, however, the down quark has no lower-generation partner, and therefore it can only decay into the up quark of the same generation.

3. Quark Decay Patterns

Although the decay patterns of the down quark and the strange quark may appear similar at first glance, the W^- boson involved in the decay behaves differently depending on the type of parent hadron (the hadron in which the quark was originally bound).

In down-quark decay, the W^- boson decays only into an electron and an electron antineutrino.

In contrast, the W^- boson produced in strange-quark decay can undergo either leptonic decay or non-leptonic (hadronic) decay, depending on the parent hadron.

In the hadronic case, the W^- boson produces mesons such as π^-.

3. Quark Decay Patterns

Leptonic decay

- K^- (*s anti-u*) : $s \rightarrow$ ***u*** W^-, $W^- \rightarrow \mu^-$ *anti-*ν_μ

Hadronic decay

- K^- (*s anti-u*) : $s \rightarrow$ ***u*** W^-, $W^- \rightarrow \pi^-$ (***d anti-u***) (the remaining *anti-u* from the parent hadron combines with the produced "*u*" to form a π^0)
- Λ^0 (u d s) : $s \rightarrow$ ***u*** W^-, $W^- \rightarrow \pi^-$ (***d anti-u***) (the produced "*u*" combines with the remaining "*u d*" quarks of the parent hadron to form a proton)

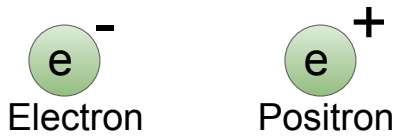
Strange-quark decays frequently produce final states containing "***u*, *anti-u*, and *d***" quarks. Since the strange quark is a down-type quark, this suggests that its effective decay structure can be regarded as involving the components "***u anti-u d***".

4. Antiparticles and Strange Quarks

An antiparticle is a particle that has the same mass and spin as its corresponding particle, but with the signs of its quantum numbers—such as electric charge—reversed.

Typical examples include the positron as the antiparticle of the electron, and the antiproton as the antiparticle of the proton.

Particles and antiparticles obey CPT symmetry.



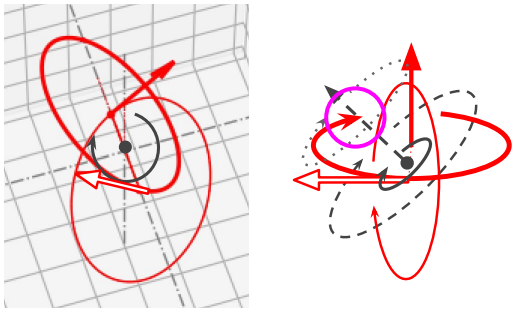
CPT refers to the combined operation of Charge Conjugation (reversing the signs of quantum numbers such as electric charge, baryon number, and lepton number), Parity (spatial inversion), and Time Reversal (reversing the direction of time).

4. Antiparticles and Strange Quarks

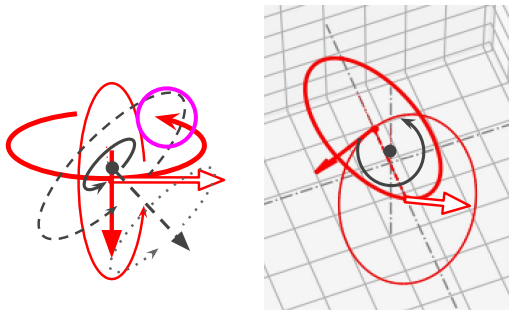
In R-space Theory, particles are represented as rotational vectors describing vortices of angular momentum, and they additionally possess oscillations (rotations) corresponding to color charge.

In this framework, the antiparticle is simply the mirror image of the particle, as following.

up quark



anti-up quark



4. Antiparticles and Strange Quarks

When a rotational vector is mirrored, the direction of rotation is reversed.

As explained in Chapter 6, rotation in R-space is equivalent to the wave equation that represents the phase of the action.

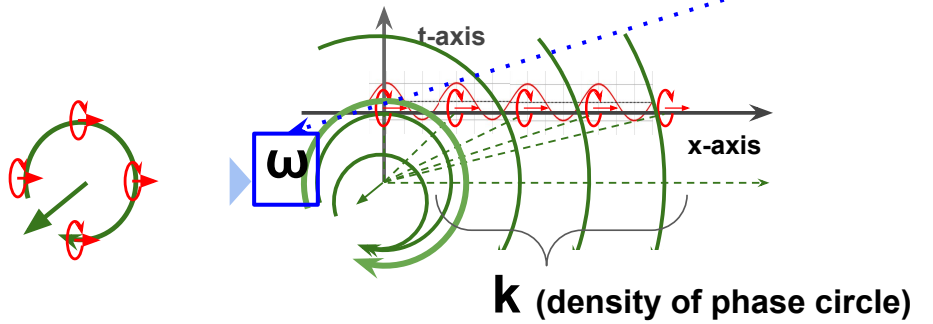
Thus, reversing the direction of rotation (i.e., flipping the sign of ω) is equivalent to reversing the time that advances the phase of the wave.

Action

$$S = p\,x - E\,t$$

Plane wave

$$\psi(x,t) = Ae^{i(kx-\omega t)}$$



4. Antiparticles and Strange Quarks

Furthermore, since electric charge in R-space Theory is defined, as described in Chapter 2, as the angular momentum of the rotational vector, reversing the direction of rotation also reverses the direction of angular momentum—and therefore the electric charge.

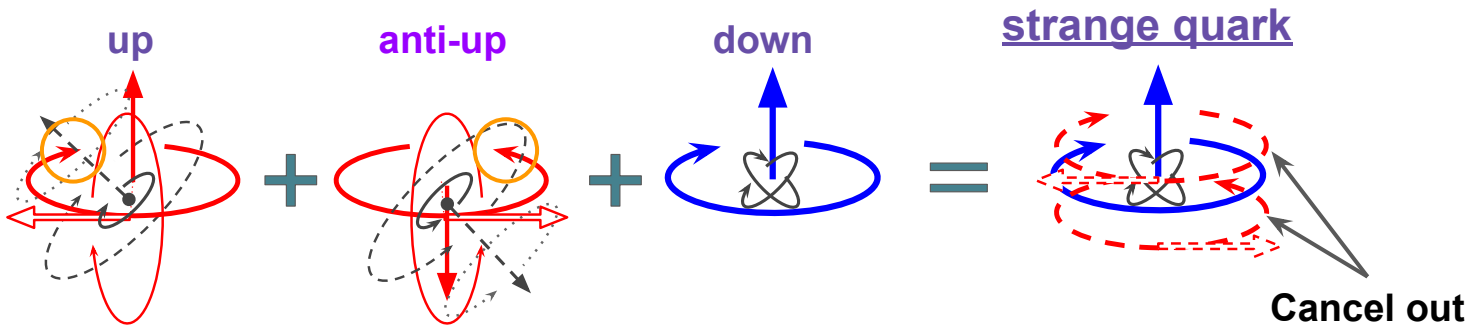
It should be noted that the arrows representing rotational vectors are drawn flipped vertically, so the figure is not a simple mirror reflection.

This is because the arrows indicate the direction of rotation: when the rotational direction is reversed by the mirror operation, the arrow itself must also be reversed to correctly represent the new rotational orientation.

4. Antiparticles and Strange Quarks

If a strange quark is assumed to contain components of "***u anti-u d***", it can be visualized as these components combined.

The "***u***" and "***anti-u***" components cancel each other out, leaving only the "***d***" component, which gives the strange quark its down-type properties.



5. The Mechanism of Quark Generations

From the relationships described above, the generation of quarks can be conceptualized as follows:

1) Rotational vector + 1st color-charge rotation → **up quark** (rot_vec + rot_c1)

2) Up quark + second color-charge rotation → **down quark** (rot_vec + rot_c1 + rot_c2)

3) Down quark + third color-charge rotation + third anti-color-charge rotation → **strange quark** (rot_vec + rot_c1 + rot_c2 + rot_c3 + anti-rot_c3)

At this point, the generation pattern changes.

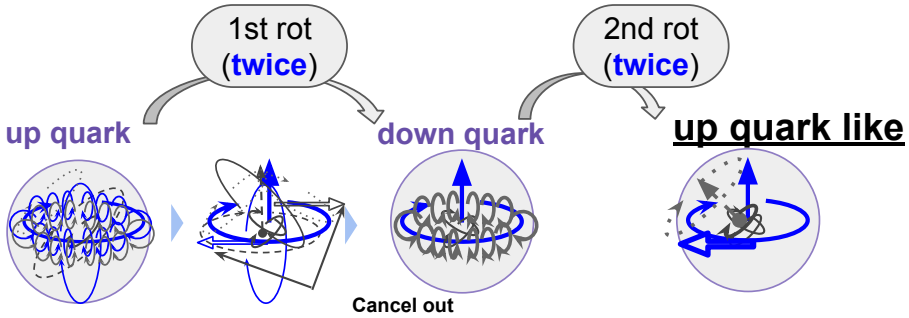
A third color-charge rotation attempts to act on the down quark.

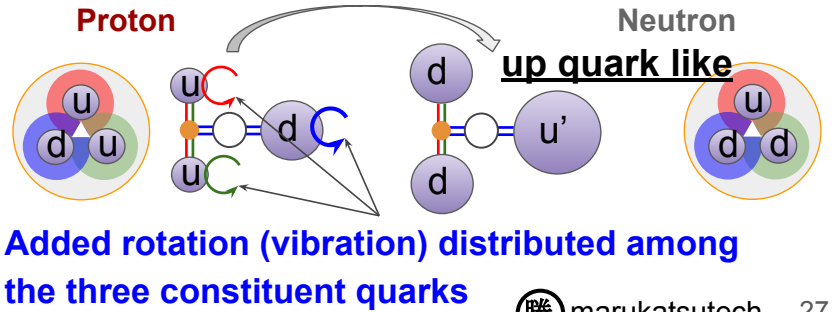
5. The Mechanism of Quark Generations

This state is identical to the up-quark-like configuration that appears during neutron formation, as explained in the previous chapter.

However, this up-quark-like state can only be generated inside a baryon, and even there it is unstable, repeatedly decaying and being reabsorbed.

This is because the added rotation consists of three components corresponding to three spatial dimensions, and within a baryon these components are distributed among the three constituent quarks.





Added rotation (vibration) distributed among the three constituent quarks

5. The Mechanism of Quark Generations

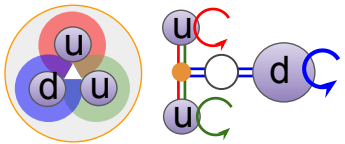
For this reason, the strange quark absorbs the additional energy through the “inclusion of an antiparticle component”—namely, a rotation corresponding to anti-color charge.

Through this rotation, the strange quark acquires an internal structure containing the components “***u anti-u d***”.

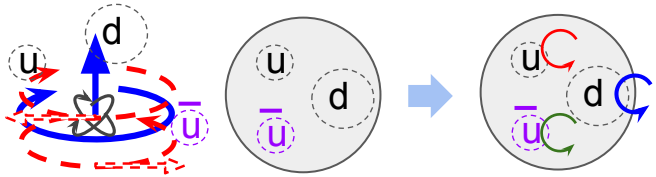
This configuration resembles that of a baryon.

Therefore, when the next rotation is added, the strange quark transforms into a charm quark.

Proton → Neutron



Strange → Charm



5. The Mechanism of Quark Generations

4) Strange quark + fourth color-charge rotation → **charm quark** (rot_vec + rot_c1 + rot_c2 + rot_c3 + anti-rot_c3 + rot_c4)

The charm quark corresponds to a state in which each of the strange-quark components "***u anti-u, d***" receives an additional color-charge rotation, resulting in a structure analogous to "***d anti-d u-like***".

5) Charm quark + fifth color-charge rotation → **bottom quark** (rot_vec + rot_c1 + rot_c2 + rot_c3 + anti-rot_c3 + rot_c4 + rot_c5)

The bottom quark is produced when an additional color-charge rotation acts on the charm quark.

Each of the charm-quark components "***d anti-d u-like***" receives a color-charge rotation.

5. The Mechanism of Quark Generations

Since there is no “higher rotation” above the u-like component, it returns to a “**d**” state, producing a structure consisting of “***u-like anti-u-like d***”.

6) Bottom quark + sixth color-charge rotation → **top quark** (rot_vec + rot_c1 + rot_c2 + rot_c3 + anti-rot_c3 + rot_c4 + rot_c5 + rot_c6)

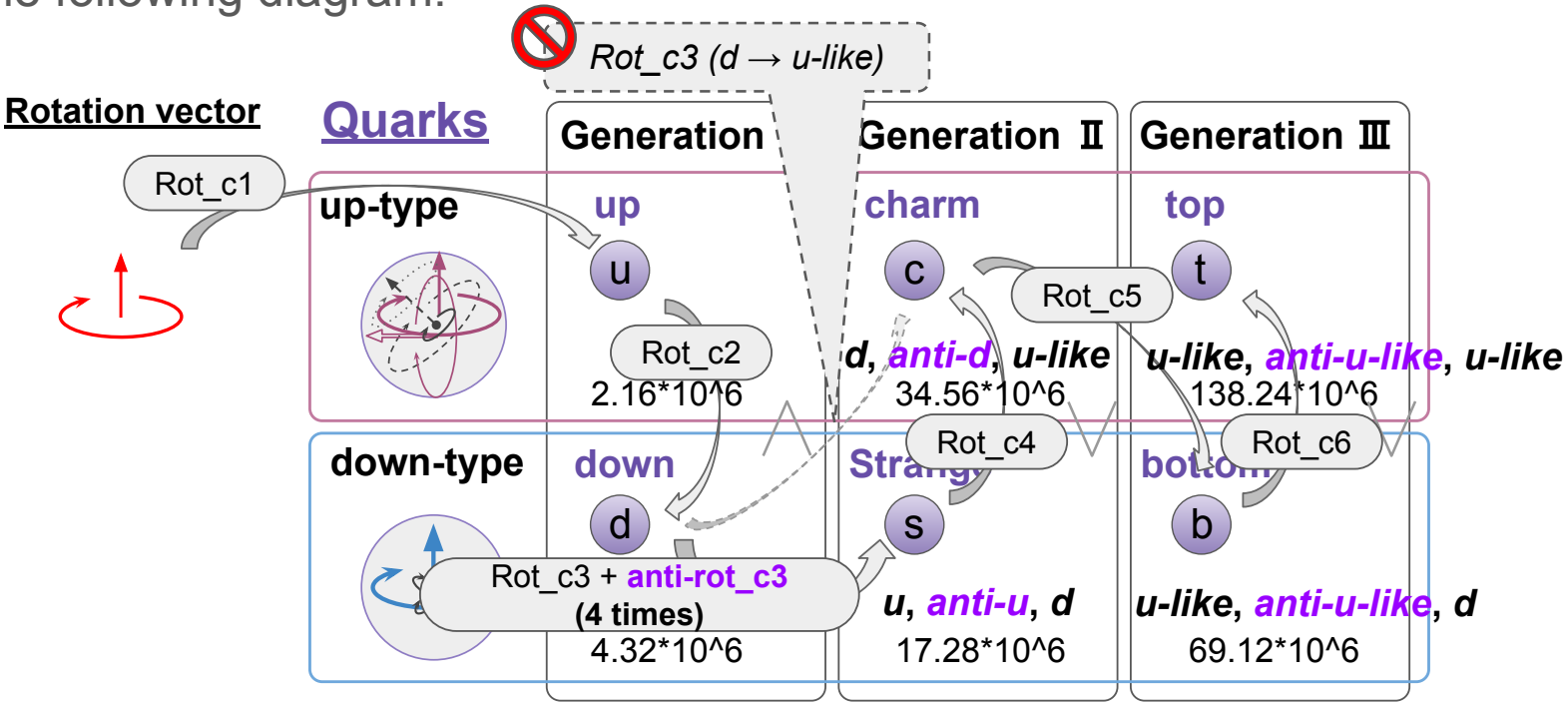
Finally, the top quark is generated when yet another color-charge rotation acts on the bottom quark.

For the bottom-quark components “***u-like anti-u-like d***”, there is again no higher rotational state above u-like.

Thus, the “*d*” component becomes u-like, resulting in a configuration of “***u-like anti-u-like u-like***”. This state is believed to be too unstable to hadronize.

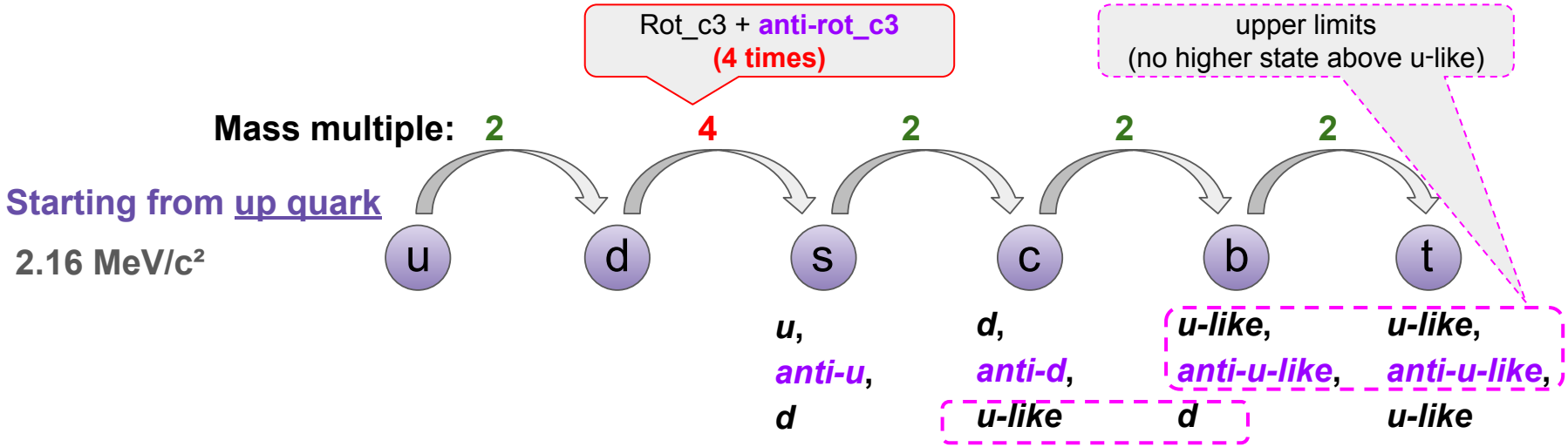
5. The Mechanism of Quark Generations

The generation pattern of quark flavors described above can be illustrated in the following diagram.



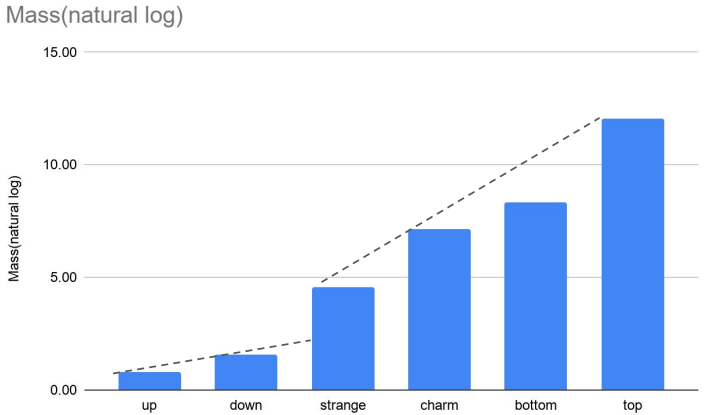
5. The Mechanism of Quark Generations

Using this mechanism to compute the masses of each flavor, one finds that the effects of antiparticle contributions (such as the additional anti-color rotation in the strange quark) and the upper limits of color-charge rotation (the absence of a higher state above u-like) appear clearly in the mass hierarchy.



5. The Mechanism of Quark Generations

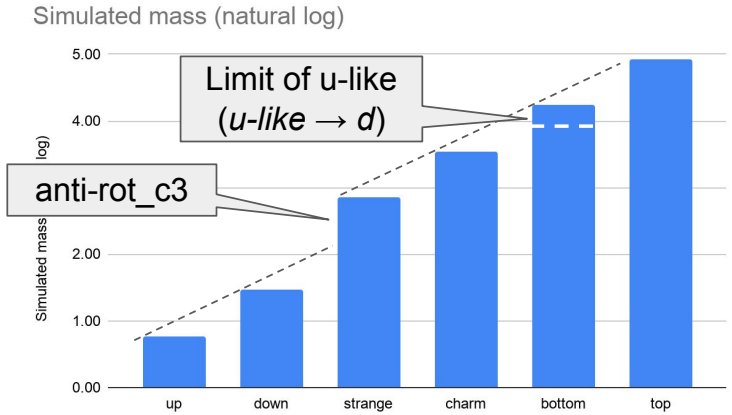
Observed



	Mass (MeV/c^2)	Mass (log10)	Mass (natural log)
up	2.16	0.33	0.77
down	4.7	0.67	1.55
strange	93.5	1.97	4.54
charm	1273	3.10	7.15
bottom	4183	3.62	8.34
top	172570	5.24	12.06

Simulated

Simulated mass = the mass of the previous flavor (previous rotational state) * **n**

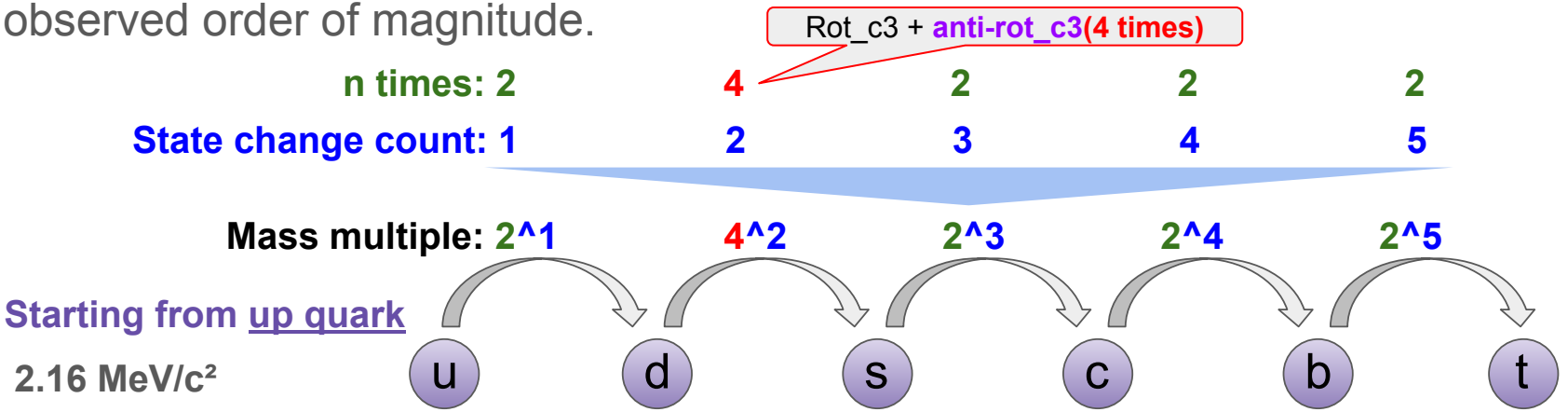


	Rotation	n times	Simulated mass (MeV/c^2)	Simulated mass (natural log)
up	c1		2.16	0.77
down	c2	2	4.32	1.46
strange	c3+anti-c3	**4**	17.28	2.85
charm	c4	2	34.56	3.54
bottom	c5	2	69.12	4.24
top	c6	2	138.24	4.93

5. The Mechanism of Quark Generations

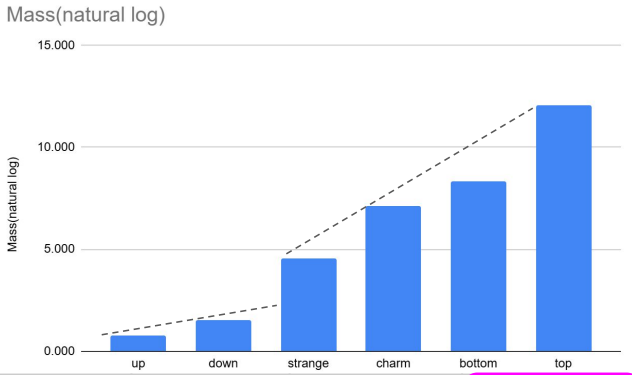
Moreover, since rotational operations are represented arithmetically as multiplications, their cumulative effect can be taken into account by treating the number of state changes produced by color-charge rotations as an exponent—expressed as “n times ^ state change count”.

With this modification, the calculated values approach the experimentally observed order of magnitude.



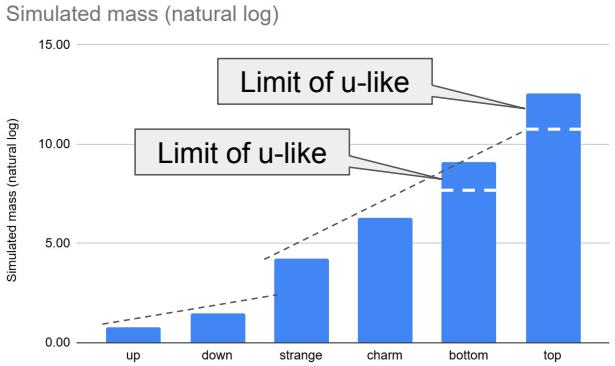
5. The Mechanism of Quark Generations

Observed



	Mass (MeV/c^2)	Mass (log10)	Mass (natural log)
up	2.16	0.33	0.77
down	4.7	0.67	1.55
strange	93.5	1.97	4.54
charm	1273	3.10	7.15
bottom	4183	3.62	8.34
top	172570	5.24	12.06

Simulated (with the cumulative effect)



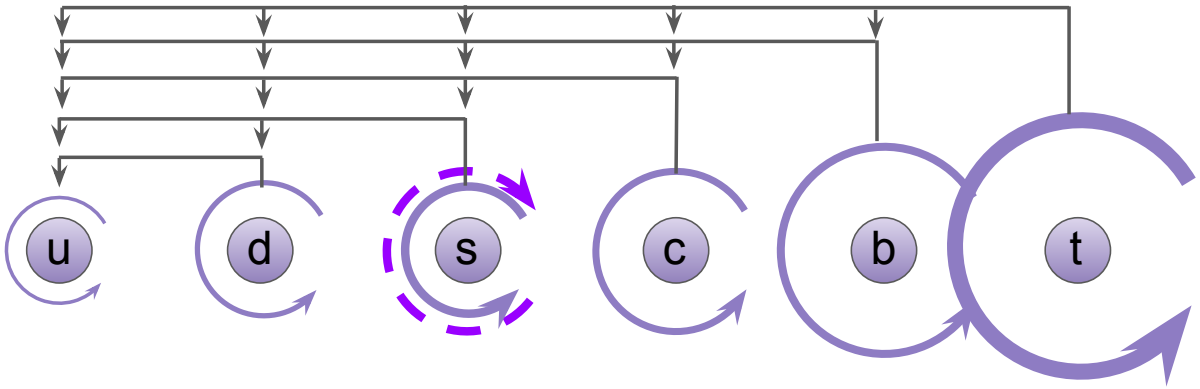
Simulated mass = the mass of the previous flavor (previous rotational state) * **n** ^ **state change count**

	Rotation	**n times**	**state change count**	Simulated mass (MeV/c^2)	Simulated mass (natural log)
up	c1			2.16	0.77
down	c2	2	1	4.32	1.46
strange	c3+anti-c3	**4**	2	69.12	4.24
charm	c4	2	3	552.96	6.32
bottom	c5	2	4	8847.36	9.09
top	c6	2	5	283115.52	12.55

5. The Mechanism of Quark Generations

The fact that the mass simulation using this exponential scaling approaches the experimentally observed order of magnitude suggests that each quark flavor possesses a hierarchical structure generated by successive rotations.

This is because, in order to rotate a given flavor state, the rotational operation must act on all of the rotational components that the flavor already contains.



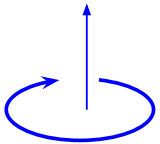
6. Relationship between Hypercharge and R-space Theory

In R-space Theory, electric charge is defined by the number of rotational vectors, as explained in the previous chapters.

$$q = (-1)^n \cdot \frac{n}{3}$$

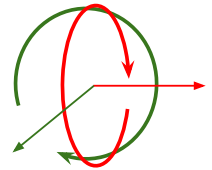
q: charge

n: number of rotation vectors



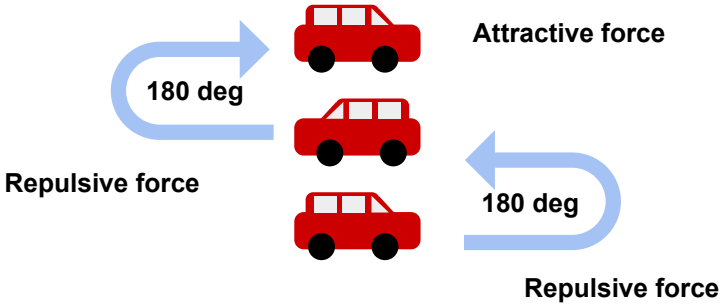
down quark

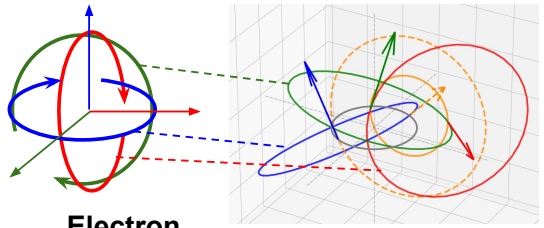
q = (-1)^**1** * **1**/3 = -1/3



up quark

q = (-1)^**2** * **2**/3 = +2/3





Electron

q = (-1)^**3** * **3**/3 = -1

6. Relationship between Hypercharge and R-space Theory

In the Standard Model, the quark charge Q is expressed using the extended Gell-Mann–Nishijima formula,

$Q = T_3 + Y_W / 2$, where T_3 is the third component of weak isospin and Y_W is the weak hypercharge.

Extended Gell-Mann–Nishijima Formula

Up-type quarks (*u, c, t*) — left-handed u_L, c_L, t_L

$T_3 = + 1/2$, $Y_W = + 1/3 \rightarrow Q = 3/6 + ⅙ = + 2/3$

Down-type quarks (*d, s, b*) — left-handed d_L, s_L, b_L

$T_3 = - 1/2$, $Y_W = + 1/3 \rightarrow Q = - 3/6 + 1/6 = - 1/3$

6. Relationship between Hypercharge and R-space Theory

In this formula, the charge (weak hypercharge) is treated as a fixed quantity.

In contrast, R-space Theory defines charge by the number of rotational vectors—yet it yields the same numerical results.

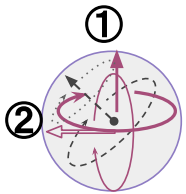
R-space Theory Charge Formula

Up-type quarks (*u, c, t*)

$(-1)^{n} = (-1)^{2} = +1$

$n / 3 = 2/3$

$Q = 1 \times \frac{2}{3} = +2/3$

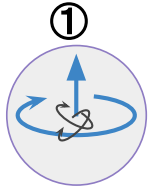


Down-type quarks (*d, s, b*)

$(-1)^{n} = (-1)^{1} = -1$

$n / 3 = 1/3$

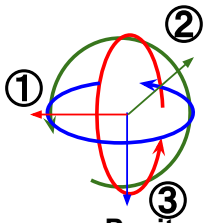
$Q = 1 \times \frac{1}{3} = -1/3$



6. Relationship between Hypercharge and R-space Theory

Interpreting the extended Gell-Mann–Nishijima formula through the R-space Theory viewpoint—where charge is defined by the number of rotational vectors—suggests that the weak isospin T_3 corresponds to one-half of the electron or positron charge, while the weak hypercharge Y_W corresponds to the charge carried by a single rotational vector, namely 1/3.

Up-type quarks (*u, c, t*)



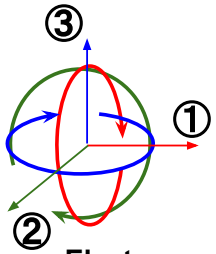
$T_3 = 1/2$, $Y_W = 1/3$

$$\frac{3/3}{2} + \frac{1/3}{2} = +2/3$$

Positron

q = (-1)^**3** * **-3**/3 = **+1 = 3/3**

Down-type quarks (*d, s, b*)



$T_3 = -1/2$, $Y_W = 1/3$

$$\frac{-3/3}{2} + \frac{1/3}{2} = -1/3$$

Electron

q = (-1)^**3** * **3**/3 = **-1 = -3/3**

Note; in standard model, positron: T_3**=0**, Y=+2 of the, electron: T_3=-1/2, Y=-1

6. Relationship between Hypercharge and R-space Theory

In the Standard Model, the weak isospin of the positron is zero.

This arises from the breaking of electroweak symmetry.

The details of this symmetry breaking will be explained in the following chapters.

In the R-space formulation, the greater the number of rotational-vector axes, the higher the interaction frequency (probability), which naturally explains the magnitude of electric charge.

Thus, quarks—including their generational structure—can be described entirely in terms of combinations of rotational vectors, representing vortices of angular momentum.

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This is all for my presentation.

Thanks for watching!

